Lemma 2.43 ([17, Lemma 9.2])**.** *Let $L_1, L_2 \subseteq \mathrm{VF}^m$ be two minimal limit sets of f at a. Then the closures of L_1, L_2 coincide.*

This lemma justifies writing $\lim_{A\to a} f = L$ when L is a closed (hence the unique) minimal limit set of f at a, in particular, when L is finite and minimal.

Let $A \subseteq \mathrm{VF}^n$ be a definable set with $\dim_{\mathrm{VF}}(A) = n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ a definable finitary function. We say that f is *continuous* if $\lim_{A\to a} f = f(a)$ for all $a \in A$.

Lemma 2.44. *Suppose $n > 0$. Then f is continuous almost everywhere.*

Proof. This can be reduced to the familiar case that f is indeed a function into VF^m as follows. By Lemma 2.12 and compactness, there are a definable set U in RV and a definable function $g : A \times \mathrm{VF}^m \longrightarrow A \times U$ such that, for all $a \in A$, $\mathrm{pr}_A(g(a, b)) = a$ and $g \restriction (a \times f(a))$ is injective. For each $t \in U$, let

$$A_t = \{a \in A \mid (a, t) \in g(a \times f(a))\}$$

and $f_t \subseteq f$ be the obvious t-definable function $A_t \longrightarrow \mathrm{VF}^m$. The discontinuity locus E of f is definable: the limit condition is first-order, and minimality of the finite set $f(a)$ can be tested by removing one of its elements at a time. Suppose for contradiction that $\dim_{\mathrm{VF}}(E) = n$. By Lemma 2.17, E contains a nonempty open polydisc. Apply generic local constancy [17, Lemma 8.9] to the finite label-set map

$$a \longmapsto g(a \times f(a))_{\mathrm{RV}}.$$

After shrinking to a nonempty open polydisc $P \subseteq E$, this set is a fixed finite set T. If T is empty, then f is the constant empty correspondence on P and is continuous there, a contradiction.

For each $t \in T$, the inverse branch f_t is defined on all of P. Extend it by zero outside A_t and apply Lemma 2.36. The finitely many branch-exception sets have union of VF-dimension less than n, so choose $a \in P$ outside their union. All branches are continuous at a. Their finite union has limit set $f(a)$; this limit set is minimal because the distinct values at a can be separated by sufficiently small discs and, since $n > 0$, the punctured polydisc still accumulates at a. Thus f is continuous at a, contradicting $P \subseteq E$. Hence $\dim_{\mathrm{VF}}(E) < n$. □

Resolved referee R039 [Gap; revised] – One continuous branch is not enough.
Original source line 1068. The argument finds one t for which A_t has full dimension and f_t is generically continuous. That alone does not account for all branches of f. Apply generic local constancy to the finite RV-label set $a \mapsto g(a \times f(a))_{\mathrm{RV}}$, then simultaneously apply generic continuity to its finitely many inverse branches on each neighborhood. To justify the global dimension conclusion, suppose the discontinuity locus had full dimension and choose an open polydisc in it where this label set is constant; the finite collection of branch-exception sets cannot fill that polydisc. The empty label set simply gives the constant empty correspondence and should be handled separately. This supplies the missing coverage argument.

Author resolution. The author approves the expanded proof and the restriction $n > 0$. The original ending uses the dimension principle that an RV-indexed definable union of sets of VF-dimension less than n still has dimension less than n. The expanded argument obtains the same coverage by making the finite label set locally constant and treating all branches together, including the empty-label case. The positive-arity hypothesis is needed for the unchanged punctured limit convention.

Unapplied R041/R042 proposal for author review. Lemma 2.48 keeps its geometric conclusion and makes its finite VF arity explicit. Its proof now treats finite missing boundary points and unbounded output separately, using finite nonempty algebraic choices and Lemma 2.45; this also justifies the temporary base enlargement. The closed-graph criterion retains the intended projective closure: the affine target remains B, while graph closure is explicitly taken in $A \times B^{\mathrm{p}}$. Its reverse proof first selects bad inputs and then takes a finite projective limit of their images. The projective-limit remark below is phrased in the

valuation topology. These green edits are proposals only; neither referee annotation is marked resolved. R166's separate wording is unchanged.

Lemma 2.45 ([17, Lemma 9.5])**.** *Suppose that $f : \mathcal{M} \smallsetminus 0 \longrightarrow \mathcal{P}(A)$ is a definable finitary function, where A is a closed and bounded set in* VF*. Then there is a definable finite set $L \subseteq A$ such that $\lim_{\mathcal{M}\smallsetminus 0 \to 0} f = L$.*

We remark that the finite-limit theorem also holds in the coordinatewise projective line $(\mathbb{P}^1(\mathrm{VF}))^m$, with the valuation topology: if $A \subseteq \mathrm{VF}^m$ is not bounded, take its closure there and allow infinite coordinates in L. This is the projective version of the same cited theorem; no topological local compactness is assumed.

revision of the next few lemmas

Lemma 2.46 (Curve selection)**.** *Suppose that $\mathbb{S}$ is* VF*-generated. Let $A \subseteq \mathrm{VF}^n$ be definable and let b be a point in its closure in the coordinatewise projective line $(\mathbb{P}^1(\mathrm{VF}))^n$, with the valuation topology. Then there is a b-definable continuous finitary function $g : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ whose values are nonempty and such that*

$$\lim_{\mathcal{M}\setminus\{0\} \longrightarrow 0} g = \{b\}.$$

Resolved referee R165 [Clarification; new] – Definability and nonempty values in curve selection.
Original source line 1081. State that A is definable. In the proof require $g(c) \neq \varnothing$ for every c (or at least on a punctured neighborhood of 0): "finitary" in the earlier definition allows empty values, and $g(c) \subseteq \mathfrak{o}(b, \mathrm{val}(c))$ alone is also satisfied by the identically empty correspondence. The algebraic-closure construction, once its base hypothesis is corrected below, supplies finite nonempty values and hence the stronger conclusion $\lim g = \{b\}$.

Author resolution. The author approves explicitly taking $A \subseteq \mathrm{VF}^n$ definable and requiring finite nonempty values at every input. The construction gives the stronger singleton limit $\{b\}$. Projective charts handle infinite coordinates without changing the affine target set of the curve.

Proof. Use a coordinatewise projective chart around b to reduce to the case $b \in \mathrm{VF}^n$: at an infinite coordinate use inversion and restrict A to the chart. These changes are definable over b and preserve convergence and continuity locally.

Naming the finite tuple b and a nonzero $c \in \mathcal{M}$ preserves VF-generation and the standing nontriviality of the ground valuation. Hence $\mathrm{acl}(\mathbb{S}, b, c)$ is a model of $\mathrm{ACVF}(0,0)$. Since $b \in \overline{A}$, the definable set $A \cap \mathfrak{o}(b, \mathrm{val}(c))$ is nonempty, so elementarity gives a point of that intersection algebraic over $(\mathbb{S}, b, c)$. Compactness supplies a b-definable finitary function g_0 on $\mathcal{M} \setminus \{0\}$ such that

$$\varnothing \neq g_0(c) \subseteq A \cap \mathfrak{o}(b, \mathrm{val}(c))$$

for every c. Thus g_0 converges to $\{b\}$, and nonempty values make this limit minimal.

By Lemma 2.44, with its one-dimensional source, g_0 is continuous outside a finite set. Choose a b-definable clopen disc $D \subseteq \mathcal{M}$ around 0 such that g_0 is continuous on $D \setminus \{0\}$. Also $\mathrm{acl}(\mathbb{S}, b)$ is a model, so A contains a finite nonempty b-definable set E. Define

$$g(c) = \begin{cases} g_0(c), & c \in D \setminus \{0\}, \\ E, & c \in \mathcal{M} \setminus D. \end{cases}$$

The two pieces are relatively clopen in $\mathcal{M} \setminus \{0\}$, so g is continuous, finite-valued and nonempty everywhere. Its limit at 0 is still $\{b\}$. Finally transport the construction back through the projective

chart. This avoids assuming that a definable value-group radius is the valuation of a definable scalar. □

Resolved referee R040 [Error] – Curve selection requires a restriction on the base.
Original source line 1088. The assertion is false for the arbitrary $\mathbb{S}$ allowed so far. Let K be a nontrivially valued algebraically closed base field, add an RV parameter t with $\mathrm{vrv}(t) = 0$ transcendental over its residue field, and put $A = \{(x, y) : y \neq 0, \mathrm{rv}(x/y) = t\}$. Then $(0, 0) \in \overline{A}$. A nonempty finite $\mathbb{S}$-definable choice in A at an input $c \in K \cap (\mathcal{M} \setminus \{0\})$ would have coordinates algebraic over K, t, c, hence in K by Lemma 2.13; their ratio cannot have leading term t. Even though finitary values may be empty, $b \in \lim g$ forces their support to accumulate at 0; by C-minimality that support contains a punctured neighborhood, whose defining radius lies in $\Gamma(K)$ since adjoining the residue parameter t does not enlarge definable values. It therefore contains such an input $c \in K$. Thus the claimed curve does not exist over that base. Impose VF-generation before this lemma (as already suggested in the human report), or allow explicitly specified extra VF representatives. Under VF-generation, the added nonzero $c \in \mathcal{M}$ makes the algebraic closure a model, and the displayed construction yields nonempty finite values converging to b itself. For the last rescaling step, first shrink to a disc whose radius is the valuation of a definable scalar; the original definable radius need not be realized by such a scalar.

Author resolution. The author approves VF-generation as a local hypothesis of this lemma, validating the algebraic-closure model argument. The continuous germ is extended by a constant finite nonempty set on a clopen complement, so no definable scalar realizing an arbitrary definable radius is assumed. Later geometric uses may temporarily enlarge the base; no global VF-generation assumption is imposed here.

This is an analogue of the highly useful o-minimal curve selection.

Remark 2.47. Although the Γ-sort is essentially an o-minimal group, curve selection fails due to the presence of the top element ∞. For example, the set $\Gamma \times (0, \infty) \subseteq \Gamma_\infty^2$ contains curves approaching the boundary point $(0, 0)$ or (∞, ∞) but no curves approaching the boundary point $(\infty, 0)$, because all curves are lines.

Referee R166 [Typo; new] – Eventually affine curves in the value group.
Original source line 1094. Replace "all curves are lines" by "every definable one-parameter curve is piecewise rational-affine". Near a one-sided endpoint, one affine piece remains; a positive affine coordinate cannot tend to 0 while another coordinate tends to ∞. This is the precise reason the stated boundary point cannot be approached.

Lemma 2.48. *Let $A \subseteq \mathrm{VF}^n$ be definable, closed and bounded, and let $f : A \longrightarrow \mathrm{VF}^m$ be definable and continuous. Then $f(A)$ is also closed and bounded.*

Proof. The conclusion is geometric, so we may temporarily enlarge the parameter base to a small VF-generated base over which all the data are definable. The standing nontriviality of the ground valuation is preserved. Over this base, and after naming any additional finite VF tuple, algebraic closure is a model of $\mathrm{ACVF}(0, 0)$. Elementarity and compactness therefore give finite nonempty definable choices in the nonempty families below. No choice over the original parameter base is asserted.

First let $b \in \overline{f(A)} \setminus f(A)$, where this closure is affine, and name b. For every $c \in \mathcal{M} \setminus \{0\}$ the set

$$\{x \in A : f(x) \in \mathfrak{o}(b, \mathrm{val}(c))\}$$

is nonempty. Applying the model argument over the enlarged base together with (b, c), and then compactness, gives a definable finitary function $F : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ such that every $F(c)$ is nonempty and $f(F(c)) \subseteq \mathfrak{o}(b, \mathrm{val}(c))$. Lemma 2.45, in its finite-dimensional form, gives a finite

minimal limit set $L \subseteq A$. This set is nonempty, since F has nonempty values arbitrarily close to 0. For every $a \in L$, minimality implies that values of F meet every neighborhood of a for inputs arbitrarily close to 0: otherwise a could be removed from L. Continuity at a and the displayed containment force $f(a) = b$, contradicting $b \notin f(A)$. Thus $f(A)$ is closed.

If $f(A)$ were unbounded, one coordinate f_i would be unbounded. For every $c \in \mathcal{M} \setminus \{0\}$ the definable set

$$\{x \in A : \mathrm{val}(f_i(x)) < -\mathrm{val}(c)\}$$

would be nonempty. The same model argument and compactness give finite nonempty choices $F(c)$ inside these sets. Again Lemma 2.45 supplies a nonempty finite minimal limit set $L \subseteq A$. Choose $a \in L$. Continuity makes f_i bounded on a neighborhood of a in A, while minimality makes $F(c)$ meet that neighborhood for arbitrarily small nonzero c. The inequality $\mathrm{val}(f_i(x)) < -\mathrm{val}(c)$ for all $x \in F(c)$ contradicts this local bound. Hence $f(A)$ is bounded as well. □

Referee R041 [Gap] – Selecting and lifting the limiting curve.
Original source line 1104. If the image is unbounded but closed in affine space, its affine frontier need not contain a point outside it. Take the frontier in the coordinatewise projective line and allow an infinite coordinate. Also the existence of algebraic points in every fiber needs justification under the chosen parameter assumptions; for the curve pullback it is enough to include the curve parameter c in the algebraic closure and apply the model argument there. Preserve a nonempty finite-valued parametrization throughout and apply the limit lemma in its multidimensional form.

Corollary 2.49. *If the function f in Lemma 2.48 is injective then it is a homeomorphism from A onto $f(A)$.*

Lemma 2.50 (The closed graph criterion for continuity)**.** *Let $A \subseteq \mathrm{VF}^n$ and $B \subseteq \mathrm{VF}^m$ be definable, with B closed in affine space. Write B^{p} for the closure of B in $(\mathbb{P}^1(\mathrm{VF}))^m$, with the valuation topology. A definable function $f : A \longrightarrow B$ is continuous if and only if its graph is closed in $A \times B^{\mathrm{p}}$. If B is bounded, then $B^{\mathrm{p}} = B$, so this is the usual relative closed-graph condition.*

Proof. The graph of a continuous map into the Hausdorff space B^{p} is closed, proving the forward implication.

Conversely, suppose the graph is closed but f is not continuous at $a \in A$. Choose a clopen projective polydisc U around the finite point $f(a)$ such that

$$D = \{x \in A : f(x) \notin U\}$$

has a in its closure. As in the preceding proof, temporarily enlarge the parameter base to a small VF-generated base containing the parameters of these data. This changes neither continuity nor graph closure. By Lemma 2.46, there is a definable continuous finitary function $C : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(D)$ with nonempty values and limit $\{a\}$.

Apply the projective version of Lemma 2.45 to the finite-valued function $c \longmapsto f(C(c))$. It has a finite minimal limit set L, which is nonempty because all the values are nonempty. Every point of L belongs to the closed set $B^{\mathrm{p}} \setminus U$: minimality makes the image meet each neighborhood of such a point for inputs arbitrarily close to 0. Choose $b' \in L$. This same property, together with $C(c) \longrightarrow \{a\}$, shows that (a, b') belongs to the closure of the graph of f in $A \times B^{\mathrm{p}}$. Closedness of the graph gives $b' = f(a)$, contrary to $b' \notin U$. This proves continuity without a global VF-generation hypothesis. □

Referee R042 [Gap; revised] – The compactified closed-graph criterion.
Original source line 1117. The parenthesis appears intended to impose the correct compactified-target hypothesis; make it explicit by taking B to be a closed definable subset of $(\mathbb{P}^1(\mathrm{VF}))^m$, and take graph

closure in $A \times B$. The example $f(0) = 0$, $f(x) = 1/x$ for $x \neq 0$ has closed graph in affine VF^2 but is discontinuous at 0: this is a counterexample only to the unbounded affine-target version, not to the intended compactified statement. For the reverse implication, apply curve selection to the set of input points x with $f(x) \notin \mathfrak{b}$, approaching a; apply the finite-limit lemma to the resulting finite-valued image in B. A limit point b' lies outside the clopen neighborhood $\mathfrak{b}$, and (a, b') lies in the graph closure. Also, "projective line with the valuation topology" is more precise than a literal topological one-point compactification; local compactness is not assumed here.

Lemma 2.51. *Let $f : A \longrightarrow B$ be a definable bijection between two open subsets of* VF*. Suppose that B is bounded. Then f is weakly concentric, that is, for every $a \in A$ there is a $b \in \mathrm{VF}$ such that, for every sufficiently large $\gamma \in \Gamma$, $f(\mathfrak{o}(a, \gamma) \smallsetminus a) = \mathfrak{o}(b, \delta) \smallsetminus b$ for some $\delta \in \Gamma$.*

Note that, compared to Lemma 2.32, the assumption of this lemma is stronger and its conclusion weaker, but the point is that we do not need a partition to achieve it.

Proof. By Lemma 2.45, for any $a \in A$, $\lim_{A \to a} f$ exists in the closure of B, which we denote by b_a. By C-minimality and o-minimality in the Γ-sort, $\mathfrak{c}(b_a, \delta) \smallsetminus \mathfrak{o}(b_a, \delta) \subseteq B$ for all sufficiently large $\delta \in \Gamma$. Also, for all sufficiently large $\gamma \in \Gamma$ there is a $\delta \in \Gamma$ such that

$$f(\mathfrak{c}(a, \gamma) \smallsetminus \mathfrak{o}(a, \gamma)) = \mathfrak{c}(b_a, \delta) \smallsetminus \mathfrak{o}(b_a, \delta)$$

The lemma follows. □

Referee R043 [Gap; revised] – The annulus equality needs a proof.
Original source line 1132. Specify that A and B each have one VF-coordinate. The finite limit is a singleton $\{b\}$ by [17, Lemma 9.4]; nonemptiness follows because the values are nonempty on a punctured neighborhood. The equality of whole thin annuli needs more explanation. Here is a repair. The valuation calculation in the proof of [17, Lemma 8.3], applied to $x \mapsto \mathrm{val}(f(x) - b)$, gives

$$\mathrm{val}(f(x) - b) = q\,\mathrm{val}(x - a) + c \qquad (q \in \mathbb{Q}_{>0})$$

on a sufficiently small punctured neighborhood: each unary polynomial has eventually monomial valuation, and the limit forces positive slope. Remove the at most one exceptional input with $f(x) = b$. By C-minimality, B contains a punctured neighborhood of b. The inverse has a unique projective limit there, necessarily a, since points $x \longrightarrow a$ give $f(x) \longrightarrow b$. Apply the same valuation calculation to the inverse. The resulting two affine maps of tail radii are inverse, so every sufficiently small target shell consists exactly of images of its corresponding source shell. Taking their unions proves the asserted punctured-disc equality. Thus the lemma is repairable; an additional assertion that $q = 1$ is unnecessary.

Definition 2.52. Let $p : A \longrightarrow \Gamma$ be a definable function. We say that p is an *$\mathfrak{o}$-partition* of A if, for every $a \in A$, the function p is constant on $\mathfrak{o}(a, p(a)) \cap A$.

This makes sense even when A has no VF-coordinates, in which case any definable function $A \longrightarrow \Gamma$ is an $\mathfrak{o}$-partition of A.

Lemma 2.53. *Let p be an $\mathfrak{o}$-partition of A. Suppose that A_{RV} is doubly bounded and A_t is closed and bounded for every $t \in A_{\mathrm{RV}}$. Then $p(A)$ is doubly bounded.*

Proof. We first handle the case that A has no RV-coordinates. Since A is closed and bounded, $p(A)$ must be bounded from below. Suppose for contradiction that $p(A)$ is not bounded from above. For each $\gamma \in \Gamma$, let $A_\gamma = \{a \in A \mid p(a) > \gamma\}$. Choose a definable continuous finitary function, that is, a curve

$$f : \mathcal{M} \smallsetminus 0 \longrightarrow \mathcal{P}\Big(\bigcup_{c \in \mathcal{M} \smallsetminus 0} c \times A_{\mathrm{val}(c)} \Big)$$